GNANAMANI COLLEGE OF TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

23AD501 – DATA AND INFORMATION SECURITY

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Course Objectives:

The main learning objective of this course is to prepare the students for:

- To understand the basics of Information Security
- To know the legal, ethical and professional issues in Information Security
- To equip the students' knowledge on digital signature
- To equip the students' knowledge on email security
- To equip the students' knowledge on web security

Unit I INTRODUCTION 9

History, what is Information Security? Critical Characteristics of Information, NSTISSC Security Model, Components of an Information System, Securing the Components, Balancing Security and Access, The SDLC, The Security SDLC.

Unit II SECURITY INVESTIGATION 9

Need for Security, Business Needs, Threats, Attacks, Legal, Ethical and Professional Issues – An Overview of Computer Security - Access Control Matrix, Policy-Security policies, Confidentiality policies, Integrity policies and Hybrid policies.

Unit III DIGITAL SIGNATURE AND AUTHENTICATION 9

Digital Signature and Authentication Schemes: Digital Signature-Digital Signature Schemes and their Variants- Digital Signature Standards-Authentication: Overview-Requirements Protocols - Applications - Kerberos X.509 Directory Services.

Unit IV E-MAIL AND IP SECURITY 9

E-mail and IP Security: Electronic mail security: Email Architecture -PGP – Operational Descriptions- Key management- Trust Model- S/MIME. IP Security: Overview-Architecture - ESP, AH Protocols IPsec Modes – Security association - Key management.

Unit V WEB SECURITY 9

Web Security: Requirements- Secure Sockets Layer- Objectives-Layers -SSL secure Communication-Protocols-Transport Level Security. Secure Electronic Transaction-Entities DS Verification-SET processing.

Total = 45 Periods

Total Periods: (L=45; T=0; P=0) 45 Periods

Course Outcomes:

Students will be able:

CO1 To understand the basics of data and information security.

CO2 To understand the legal, ethical and professional issues in information security.

CO3 To understand the various authentication schemes to simulate different applications.

CO4 To understand various security practices and system security standards.

CO5 To understand the Web security protocols for E-Commerce applications.

CO's vs PO's & PSO's Mapping:

CO's \ PO's & PSO's	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	3	1	-	-	-	-	1	3	1	2	3	1
CO2	1	3	3	3	2	-	-	-	1	2	2	2	1	2
CO3	2	3	3	3	1	-	-	-	1	3	1	2	1	2
CO4	3	3	1	1	1	-	-	-	3	1	1	3	2	3
CO5	3	2	2	3	2	-	-	-	1	2	1	2	2	2
Average	**2.4**	**2.6**	**2.4**	**2.2**	**1.5**	**-**	**-**	**-**	**1.4**	**2.2**	**1.2**	**2.2**	**1.8**	**2**

Text Books:

1. Michael E Whitman and Herbert J Mattord, "Principles of Information Security, Course Technology, 6th Edition, 2017.

2. Stallings William. Cryptography and Network Security: Principles and Practice, Seventh Edition, Pearson Education, 2017.

References:

1.Harold F. Tipton, Micki Krause Nozaki,, "Information Security Management Handbook, Volume 6, 6th Edition, 2016.

2. Stuart McClure, Joel Scrambray, George Kurtz, "Hacking Exposed", McGraw- Hill, Seventh Edition, 2012.

3. Matt Bishop, "Computer Security Art and Science, Addison Wesley Reprint Edition, 2015.

4. Behrouz A Forouzan, Debdeep Mukhopadhyay, Cryptography And network security, 3rd Edition, McGraw-Hill Education, 2015.